

Human Pose Analysis

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Abstract

Human pose estimation in the OR has largely focused on detecting and localizing individual clinicians within single frames, leaving OR team dynamics across the course of a procedure comparatively unexplored. In this work, we investigate whether unsupervised clustering of pose and spatial-configuration data can reveal underlying patterns in surgical team behavior, using the MVOR dataset, the first publicly available multi-view RGB-D dataset captured during real clinical interventions. Since MVOR provides no procedure-type or team-behavior annotations, we manually organize its 1061 3D pose annotations into 24 usable operations and construct a 64-dimensional per-frame representation spanning three facets: individual pose (Geometric Pose Descriptor), interperson interaction (pose diversity and pairwise clinician distance), and group-level spatial configuration relative to the operating table. Operations, treated as multivariate time series, are compared using normalized Dynamic Time Warping (nDTW) and clustered with hierarchical clustering and k -means with DTW Barycenter Averaging (DBA); spectral clustering is evaluated but ultimately discarded after experiments indicate the data lacks non-convex structure. Since no ground-truth procedure labels exist, we validate cluster assignments against manual annotations consisting of C-arm presence and patient orientation, measured via the Adjusted Rand Index (ARI). While clustering on the full feature set yields only weak agreement with annotations, ablation studies reveal that the interperson feature group alone drives the strongest and most consistent signal, achieving an ARI of 0.55 against C-arm presence once short, low-information operations are filtered out. We further show that the drivers of this signal shift unexpectedly between pose-diversity and clinician distance subfeatures depending on the operation-length filter applied. These findings suggest that interperson spatial dynamics, rather than individual pose or global room configuration, carry the most discriminative information about surgical activity, while also highlighting the data and annotation limitations that constrain more definitive conclusions.

1 Introduction

Human pose estimation has been intensively researched in the surgical domain, motivated by its potential to support operating room activities. Within this realm, the MVOR dataset provides the first publicly available multi-view RGB-D pose dataset captured during real clinical interventions. This work and others alike focus on individual person detection methods or modelling the surgical scene at the level of a single frame. Comparatively little study has been done with regards to operating room team dynamics, how the pose configuration of clinicians evolves over the course of a procedure, and how this relates to surgical activity. In light of this gap, using the MVOR dataset, our work seeks to investigate operating room team strategy through unsupervised learning, clustering team dynamics across entire operations to determine whether any underlying patterns

emerge. Below, we present the dataset, our operation representation methodology, and our results, along with visualization of the clustering outcomes.

2 Literary Review

This section covers all the prerequisite information for the analysis that follows. We begin with a brief overview of the underlying machine learning concepts, followed by a review of relevant literature.

2.1 Machine Learning Basics

Unsupervised Machine Learning. Unlike supervised learning, which learns a mapping from inputs to known output labels, unsupervised learning trains on unlabelled data, letting the algorithm uncover hidden patterns or structure without human guidance [1]. Since the MVOR dataset contains no procedure annotations, this characteristic is what we aim to leverage in our work.

Clustering. Clustering is a class of unsupervised learning that groups similar data points together based on similarity or distance. The goal is to partition the data such that points within the same group are more similar to each other than to points in other groups [1]. Throughout this work, we make use of three clustering algorithms:

1. **Hierarchical Clustering (HC):** a recursive clustering algorithm that builds a tree-like hierarchy of clusters. We use the agglomerative (bottom-up) variant, which begins with every data point in its own cluster and recursively merges the two closest clusters until a single cluster remains. This process can be visualized using a dendrogram, and the resulting hierarchy can be cut at any level to obtain a specific number of clusters. HC requires no pre-specified number of clusters at run time, and its dendrogram structure additionally allows us to visually inspect how clusters merge [2].
2. **K-means:** a centroid-based clustering algorithm that iteratively assigns data points to the nearest centroid and then recalculates each centroid as the mean of its assigned points, repeating until convergence. Unlike HC, K-means requires the number of clusters (k) to be specified in advance [2].
3. **Spectral Clustering:** groups data based on their similarity rather than raw spatial distance, making it well suited to complex, non-convex data structures where centroid-based methods like K-means tend to fail [2].

Cluster evaluation metrics. Given our task is unsupervised, evaluating whether a clustering is meaningful requires comparing it either against a second clustering (to assess agreement between methods) or against manually annotated ground truth (to assess correspondence with real-world structure). The metric we use for this purpose is the Adjusted Rand Index (ARI).

ARI measures the similarity between two clusterings by examining every possible pair of data points and checking whether the two clusterings agree on whether that pair belongs in the same

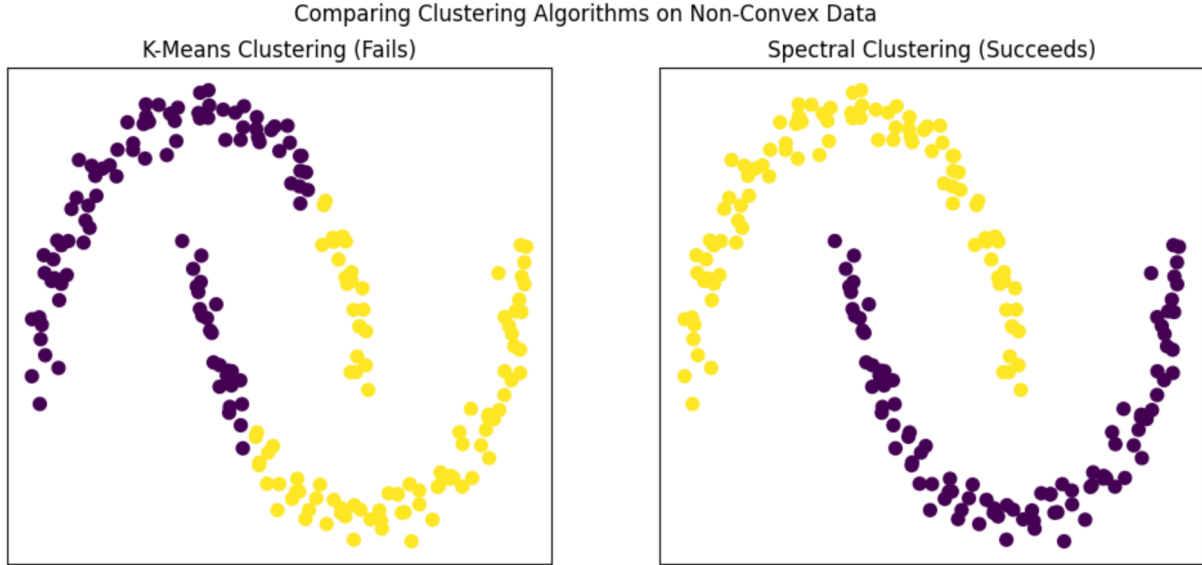


Figure 1: [K-means](#) vs. [spectral clustering](#).

cluster or apart. The raw count of agreements, however, is inflated by chance agreement that would occur even under random labelling. ARI corrects for this by adjusting the raw agreement rate against the level of agreement expected under a random model, yielding a score of 1 for perfect agreement, 0 for agreement no better than chance, and negative values for agreement worse than chance [3].

2.2 Relevant Works

Human pose estimation in the operating room presents unique challenges such as visual clutter, occlusion, similar clothing colors, and variable lighting. Early work by Kadkhodamohammadi et al. addressed this in [4] by incorporating depth information to better separate closely-positioned individuals. The same authors then extended this to a multi-view setting, fusing detections across multiple cameras to improve accuracy [5]. Further work by Kadkhodamohammadi & Padoy tackled making multi-view methods easier to deploy across different camera setups. This is done by separating tasks into two steps: 2D pose estimation, then regressing those 2D detections into a 3D pose. This framework eliminates the necessity of training data from the target environment, since the 3D regression step can be trained entirely on synthetic poses instead [6].

The culmination of these works is the MVOR dataset, which uses the methodologies developed across the preceding papers [7].

3 MVOR Dataset

3.1 Data

The MVOR dataset is made up of 732 synchronized multi-view frames from three ceiling-mounted RGB-D cameras captured across four days of procedure recordings. These multi-view frames provide 4699 bounding boxes, 2926 2D pose annotations, and 1061 3D pose annotations across 10 body parts [7].¹ Our work focuses specifically on the 1061 3D pose annotations. A sample image of the 2D and 3D annotations is shown below in Fig. 2.

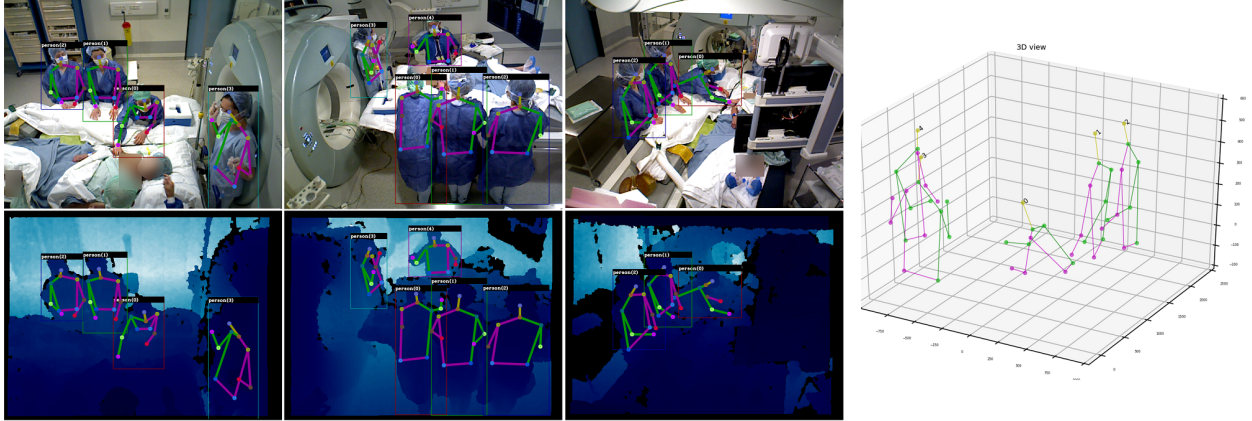


Figure 2: Sample image of the 2D and 3D annotations [7].

3.2 Operation Annotation Process

This dataset provides individual clinician keypoints; however, our task requires annotations organized across full operations. Therefore, we manually parsed the raw keypoints into frames and grouped these frames into separate operations according to the following criteria:

1. **Frame definition:** A frame is defined as the set of all keypoint annotations that are captured on the same day with the same image id.
2. **Operation definition:** An operation is defined as the span of consecutive frames between one empty frame up to but not including the next. This grouping was done leveraging the raw images.

Applying this process produced 26 total operations. Further accounting for two operations consisting of only two frames, 24 total operations remained.

¹Actual annotations contained 12 instead of 10 body parts, the extra two corresponding to the left and right hands.

4 Methodology

Although individual keypoint annotations have been organized into frames and operations, raw keypoints are not sufficient for frame representation. Hence, in the following section we go over the methodology used for frame representation, operation comparison, and cluster result validation, as well as the clustering algorithms used.

4.1 Frame Representation

4.1.1 Feature Set

While group-level pose analysis is not as established in the OR context, it has been discussed in other fields. For instance, in computer vision research, Zhang et al. modelled human group behaviour across individual, pair, and group levels to recognize human group behavior [8]. This approach motivates our operation representation. Rather than capturing team dynamics as a flat feature space, we opt to construct three feature groups modelling the various facets of group dynamics: individual behaviour, interperson interaction, and group-level context.

Individual features. In order to encode individual person configuration, we used the Geometric Pose Descriptor (GPD) proposed by Chen et al., which represents pose through the geometric relationships between joints, lines (vectors connecting two joints), and planes (formed by limbs and torso) [9]. Figure 3 illustrates an example of the joints, lines and planes that GPD captures; we adapt this representation to our dataset². GPD captures the relationships between the joints, lines and planes using eight static feature types and one temporal feature:

- Joint coordinate (j_c)
- Joint–joint distance (jj_d)
- Joint–joint orientation (jj_o)
- Joint–line distance (jl_d)
- Line–line angle (ll_a)
- Joint–plane distance (jp_d)
- Plane–plane angle (pp_a)
- Joint kinetics (**dropped due to incompatibility with dataset**)

Combining the joints, lines, and planes³ with the eight static feature types, 593 features are generated, and the total dimension of the feature set is 749, as summarized in Table 1. Given the dimension of the GPD feature set compared to our sample size (24 operations), we reduce dimensionality using principal component analysis. Applying this, we manage to reduce the GPD feature set to 50 dimensions while retaining 90% of the explained variance.

²In this paper, positions are expressed with respect to the mid-hip; thus, when computing GPD vectors, keypoint annotations were first translated to hip-centered coordinates.

³Extensive list of derived joints, lines, and planes can be found in Appendix A

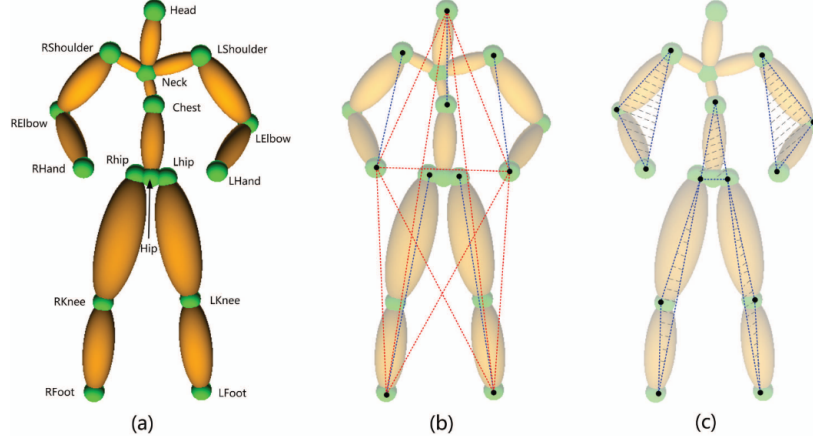


Fig. 1. (a) Skeleton model. Green spheres represent joints and orange ellipsoids represent limbs. (b) Lines. Limbs are already lines. Blue and red dash lines are additional lines of two types. (c) Planes.

Figure 3: GPD Joints, lines and plane [9]

Table 1: Summary of GPD features.

Type	j_c	jj_d	jj_o	jl_d	ll_a	jp_d	lp_a	pp_a	Total
Count	12	66	66	190	171	28	57	3	593
Dimension	36	66	198	190	171	28	57	3	749

A key concern for this representation of individual poses is that averaging GPD across all clinicians within a frame risks washing out individual pose information. However, as shown in Table 2, 536 of 626 frames consist of two or fewer clinicians; thus, averaging is unlikely to significantly dilute the pose signal.

Table 2: Clinician frame distribution.

Number of clinicians	Frame count
1 clinician	313
2 clinicians	223
3 clinicians	69
4 clinicians	15
5 clinicians	5
7 clinicians	1

Interperson features. The interperson feature group is divided into two subsets, each capturing a different aspect of interperson pose similarity.

The first subset, *perceptual pose diversity*, encapsulates the overall similarity or dissimilarity of the poses present within a frame. This feature subgroup leverages Harada et al.’s work, which presents a quantitative pose similarity evaluation metric that emulates human intuition and perception. Their key insight is that human perception of pose similarity does not scale linearly with

raw joint distances. Small differences between joints are noticeable, but beyond a certain threshold, poses are judged as equally dissimilar. Furthermore, they find that certain joints are more important than others for distinguishing poses; notably the farther a joint is from the hip, the more it contributes to distinguishing poses. To capture these findings, the paper introduces body-part weights representing the importance humans place on different joints when distinguishing poses, along with an upper threshold that caps the contribution any single joint can make to the overall distance [10].⁴

The second subset, *pairwise person distances*, captures the proximity between clinicians. Motivated by Yun et al.’s method, we compute the distance between all pairs of joints across the two individuals, then average these joint distances to get a rough estimate of the overall proximity between the clinicians [11].

These two subsets are computed across all pairs of clinicians in a frame, and the resulting pairwise values are summarized into frame-level statistics. The interperson feature set is as follows:

Pose Diversity:

- Mean pose difference
- Pose difference standard deviation
- Maximum pose difference
- Maximum pose deviation ($\max - \text{mean}$)
- Minimum pose difference

Person Distances:

- Mean clinician distance
- Clinician distance standard deviation
- Minimum clinician distance
- Maximum clinician distance

A critical limitation of this feature group is that it requires frames to have at least two clinicians present to be informative. With fewer than two clinicians, the whole feature group degrades to zero. As shown in Table 2, this is the case for around half of the frames in our dataset. Nevertheless, the other two feature groups should be able to compensate where interperson features lack. In this sense, the interperson and GPD groups are inherently complementary: interperson features are able to support frames where GPD’s averaging may lose information, and GPD supports frames where interperson features degrade to zero.

Group features. Group features capture the operating room as a whole, describing its spatial configuration. Given the OR setting, the most relevant spatial detail is how clinicians are positioned and spread relative to the operating table. In light of this, we introduce the Table Feature group, which consists of the following:

- Number of clinicians

⁴Note: Since we do not have fingertip keypoints which is emphasized in this work, we use hand keypoints as a proxy.

- Mean clinician distance from table
- Minimum clinician distance from table
- Left–right clinician distribution around table
- Number of active clinicians ⁵

Combining all feature groups results in a 64-dimensional feature space per frame. Table 3 below summarizes the full feature set.

Table 3: Full feature set.

Pose level	Feature group	Dimension	Total Dimension
Individual	GPD	50	64
Interperson	Pose diversity	5	
Interperson	Joint distances	4	
Group	Table proximity	5	

4.1.2 Feature Weighing

Looking at Table 3, it is evident that there is a disparity in the dimensionality of the different feature groups. Furthermore, while the values are all scalar, they are semantically incompatible. Left unaddressed, these issues would negatively affect cluster results; thus, we must weigh our feature groups adequately to ensure each contributes proportionally.

Generally, there are two main approaches to feature weighing: automatic weight learning and manual weighing. While automatic weight learning is widely considered to be the most optimal weighing approach, most methods obtain weights by refining them throughout the clustering process. This means that the weights obtained are cluster-based rather than globally applicable. For hierarchical clustering, this is highly incompatible since cluster assignments are not known until the end of the process. Even in the case of K-means, we are clustering 24 operations, meaning we would have to learn weights from very few sequences. For these reasons, automatic weight learning methods are incompatible for our application.

Conversely, manual weighing approaches also present their own challenges due to their subjective nature. One way to circumvent this issue entirely is to assume that all groups have equal importance and weigh them so that each contributes equally.

Another way to mitigate the subjectiveness of the weights could be through using domain knowledge; however, this requires extensive research and/or access to numerous experts. Furthermore, even with access to such resources, one would still need to find a way to reliably consolidate results into concrete feature weights.

A structured framework for generating these weights is the Analytic Hierarchy Process (AHP), a method that quantifies subjective human judgement through a logical, step-by-step protocol. In AHP, pairwise judgements are made about the relative importance of feature groups, converted into numerical values, and organized into a matrix. This matrix then aggregates all pairwise relative

⁵An active clinician is defined as one allowed within the sterile field (30 cm from the table) [12].

judgments into consistent absolute weights. Furthermore, AHP includes built-in consistency checks that flag contradictory judgements [13].

In all, given our lack of access to surgical domain experts and limited domain knowledge specific to our application, we opt not to use AHP at this stage. Instead, we simply adopt a safer default of equal weighing across all three feature groups.

4.2 Operation Comparison

Our operations consist of multiple sequential 64-dimensional vectors, making operation comparisons a multivariate time series problem. To this end, we employ Dynamic Time Warping (DTW), one of the standard distance metrics for time series data. DTW finds the optimal alignment between two time series, then computes the distance between them as a function of the total cost along the warping path [14]. Because of this, DTW distance is inherently larger for longer sequences, regardless of actual similarity. This bias is especially problematic given the large range of operation lengths, ranging from 5 to 105 frames. Figure 5 illustrates this effect, showing how frame length influences DTW with hierarchical clustering.

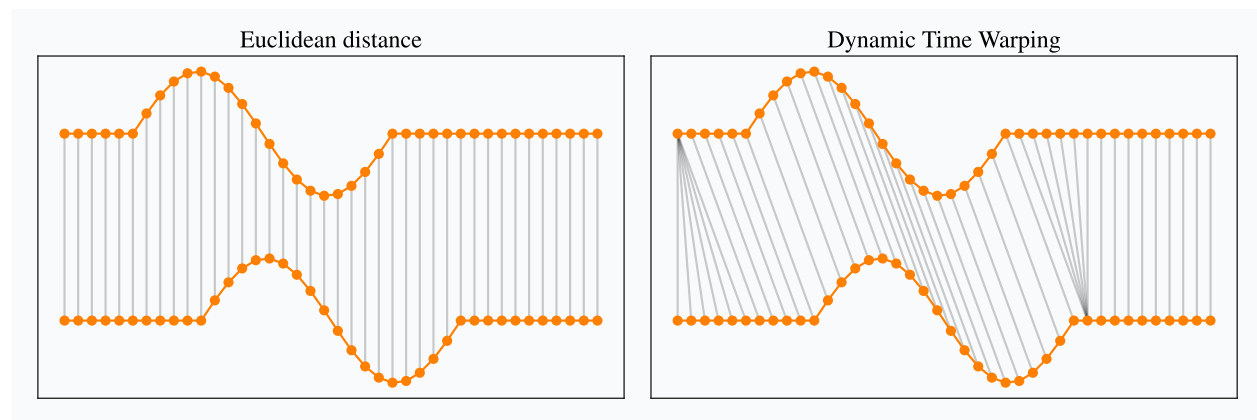


Figure 4: Euclidean Distance vs DTW [15]

To reduce the effect of varying operation length, we normalize DTW distance by the length of the warping path, resulting in normalized DTW (nDTW).

4.3 Clustering Algorithms

In this section, we discuss the clustering algorithms used throughout our analysis.

Hierarchical clustering with nDTW distance (HC). While we use HC as an actual algorithm in our analysis, it also serves an exploratory role to uncover the underlying structure of the data and identify a viable number of clusters. The resulting cluster count informs viable choices of k for the other algorithms.

K-means with DBA (DTW Barycenter Averaging). A standard method for clustering time series data. Standard K-means relies on arithmetic mean to compute cluster centroids which breaks

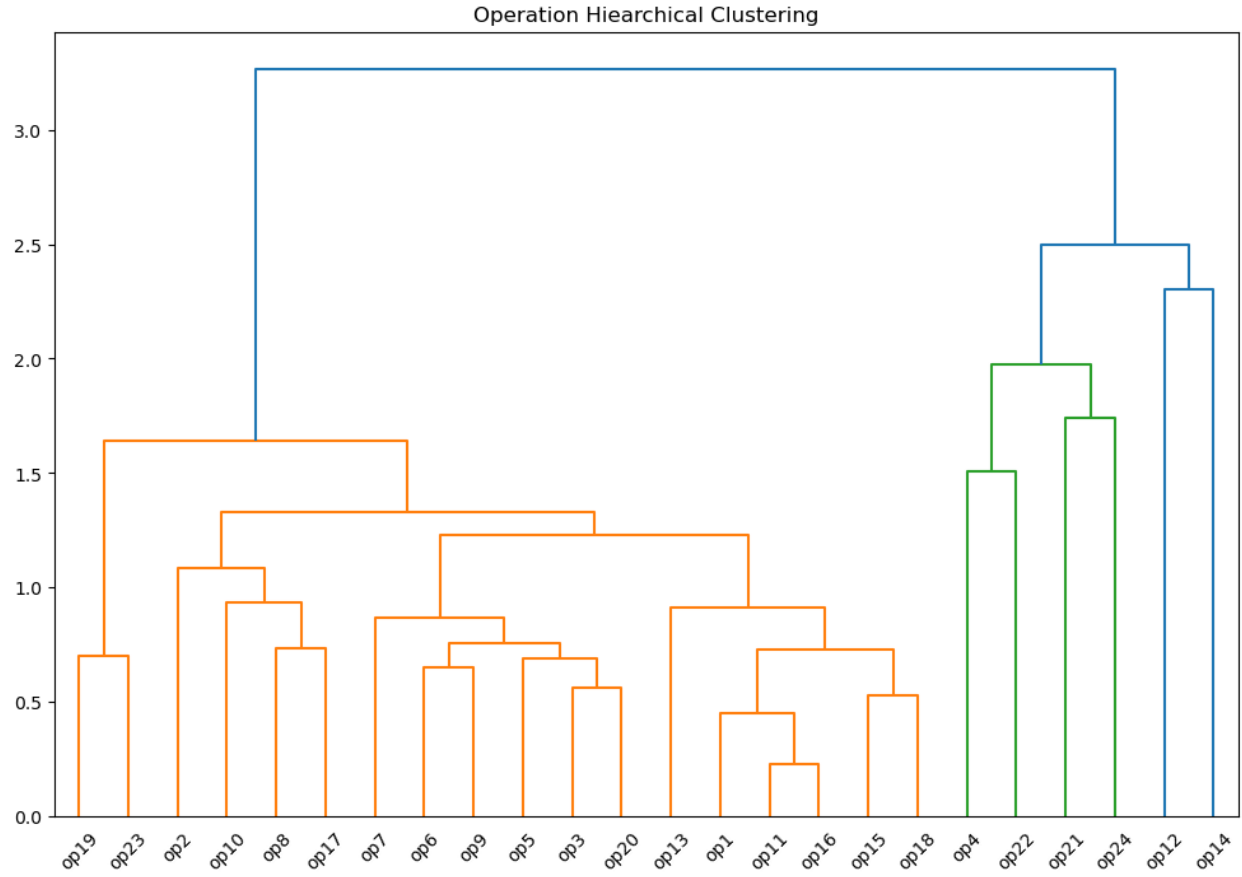


Figure 5: Blue cluster has average frame length of 101, while green cluster has average frame length of 49. Compared to the global average of 24 frames, it is evident that these outlier clusters are caused by frame length.

down for time series data. K-means with DBA address this by replacing centroid computation with DTW Barycenter Averaging.

DBA computes the average of a set of sequences through the following process. It begins with an initial average sequence C , typically initialized by randomly selecting an actual sequence from the dataset. DTW is then run between C and every sequence currently assigned to the cluster, producing an optimal alignment between C and each sequence. Once these alignments are computed, C is updated: for each point in C , its new value is set to the mean of all the points from every sequence that aligned to it under DTW, effectively producing a barycenter (a DTW-informed average) across all cluster members rather than a simple positional average. These steps are repeated until convergence at which point the barycenter C is taken to be the cluster’s centroid [16].

Since this algorithm uses DTW internally, it suffers from the same length issue discussed in the previous section. Thus, we implement K-means with normalized DBA (nDTW Barycenter Averaging), which we simply refer to as K-means going forward.

Spectral clustering. Used in the event that our data exhibits a non-convex structure that the previous two methods would fail to capture.

4.4 Validation Methods

Given the unsupervised nature of our task and the lack of annotation in the dataset, it is difficult to quantitatively validate our results. To address this, we manually annotated operations to assess whether cluster assignments correlated with any operation characteristics. While operation images did not exhibit many distinguishing features, we identified two characteristics in particular that could be annotated reliably: the presence of a C-arm machine, and the orientation the patient was lying in. Figure 7 shows how clusters look with manual labels.

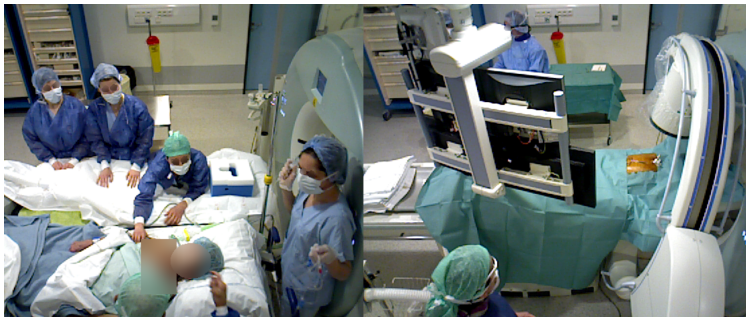


Figure 6: C-arm operation vs. no C-arm.

To measure the agreement between our clustering results and operation annotations, we use Adjusted Rand Index (ARI).

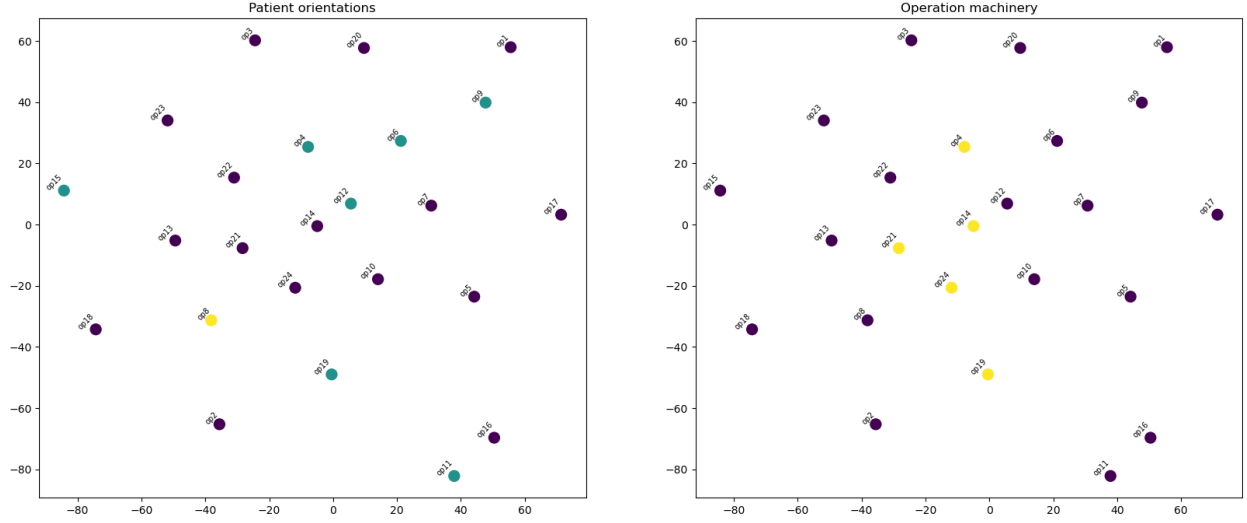


Figure 7: Plot of machinery and orientation labels.

5 Results

5.1 Initial Results

Hierarchical clustering produced two main clusters with one outlier, suggesting an optimal cluster count of $k = 2$ or $k = 3$. The outlier is operation 23, which also happens to be the second-shortest operation.

Visually, a similar pattern appears in HC ($k = 2$ & 3), spectral clustering ($k = 2$), and K-means ($k = 2$ & 3), where one cluster is in the middle and others are around it. However, this pattern is absent for spectral clustering ($k = 3$), which appears closer to random. Visualizations of these results can be found in figure 8

Despite this apparent visual consistency, ARI across algorithm pairs is no better than random for most pairs, other than K-means ($k = 2$) and HC ($k = 3$), which have slight agreement (ARI = 0.18).

Comparing cluster assignments to the annotated labels, most algorithms exhibit assignment no better than chance, with K-means ($k = 2$) being the exception, exhibiting slight agreement.

Table 4: Initial clustering ARI. Cluster labels with $k = 2$ are compared to machinery annotations, while those with $k = 3$ are compared to orientation annotations.

Algorithm	Agreement with annotations (ARI)
HC ($k = 3$)	-0.055
K-means ($k = 2$)	0.226
K-means ($k = 3$)	-0.055
Spectral Clustering ($k = 2$)	0.082
Spectral Clustering ($k = 3$)	-0.020

The results above raise the concern that we may be clustering noise rather than operation-

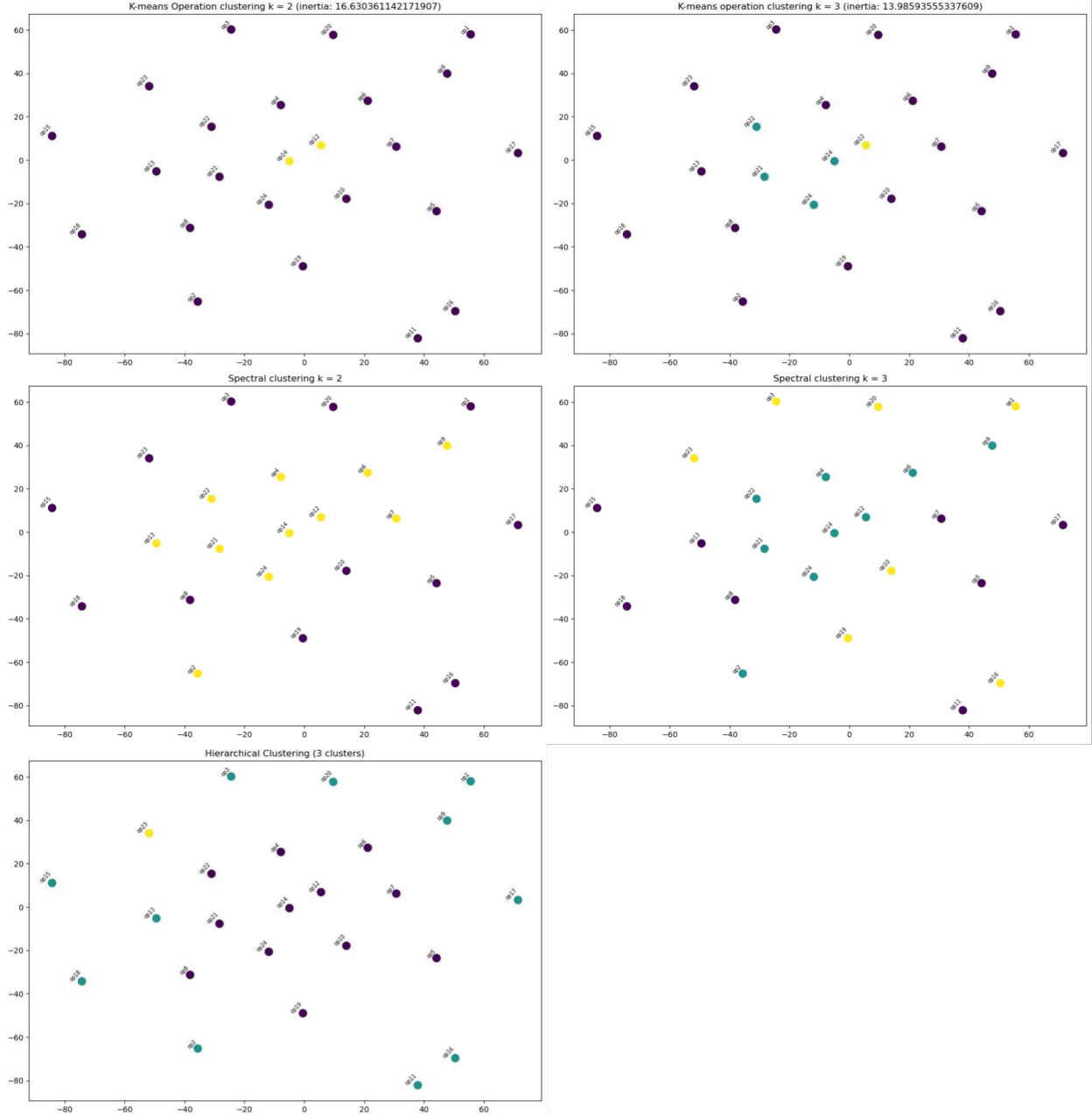


Figure 8: Initial cluster results (plots of all five cluster results).

related structure. Hence, in the section below, we investigate this possibility.

5.2 Further Experimenting

To evaluate whether clustering results reflect genuine structure, we undergo two different perturbation experiments. In each, we introduce controlled perturbations in the data to see whether cluster assignments respond accordingly.

5.2.1 90–90 Experiment

In this experiment, we create two different operation subsets: the first with the first 10% of frames removed from each operation, and the second with the last 10% of frames removed from each operation. The aim of this experiment is to determine whether both subsets yield exactly the same clustering, or whether they differ due to the perturbation.

Hierarchical clustering produced identical results at both $k = 2$ and $k = 3$. Spectral clustering was similarly consistent at $k = 2$, but became inconsistent at $k = 3$, with an ARI of 0.39. Conversely, K-means showed moderate agreement at $k = 2$ (ARI = 0.68), but this agreement dropped sharply at $k = 3$, falling to an ARI of only 0.18.

Initial results indicate that the outlier (Op23) is greatly affecting results: in cases where clustering was exactly the same, this was because the outlier was consistently assigned its own cluster while everything else fell into the remaining cluster(s). To verify this hypothesis, we removed the outlier and re-ran the comparison.

For hierarchical clustering and K-means, removing the outlier (which was artificially inflating agreement) worsened results. The same cannot be said for spectral clustering, whose agreement improved, indicating that it may not be capturing the same signal as the other two methods.

Overall, these results show clustering that is different yet slightly similar across subsets, which is consistent with the perturbations introduced.

5.2.2 Chunking Experiment

In this experiment, we take all operations with over 20 frames and chunk them into groups of 10 frames, then cluster on these chunks. The aim of this is to see whether chunks from the same operation tend to group together, or whether similar phases from different operations cluster together.

Hierarchical clustering produced five clusters, with consecutive chunks from the same operation tending to fall in the same cluster. As for spectral clustering and K-means, results look random. When comparing cluster assignments against chunk grouping per operation, agreement is no better than chance, illustrating that chunks are not clustered strictly by operation. Although operations did not cluster together, the fact that consecutive chunks cluster together is still a positive result.

Given the results thus far, it is evident that the data does not have an underlying non-convex structure; therefore, we deem spectral clustering unnecessary. Thus, we only use hierarchical clustering and K-means in subsequent analysis.

In sum, the two experiments above confirm that our algorithms are picking up some sort of signal rather than pure noise. In the following section, we perform a series of ablation studies to uncover where this signal stems from.

5.3 Ablations

In this section, we perform a series of ablation studies to uncover the origin of our underlying signal.

5.3.1 Filtering Short Operations

We suspected that shorter operations were not long enough to extract meaningful signal. Therefore, we removed shorter, noisy operations and clustered on the remaining ones. The cutoff chosen for this filtering was all operations above 10 frames, which reduces the operation count to 17.

Hierarchical clustering produced three main clusters, with dendrogram splits being far more distinct and no clear outliers.

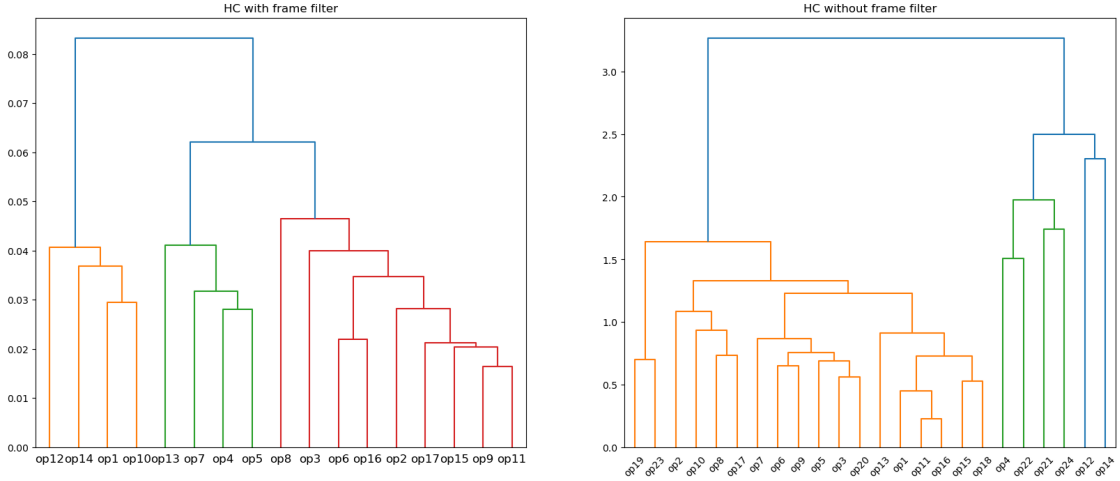


Figure 9: HC dendrogram over 10 frames vs. original dendrogram.

Visually, K-means ($k = 3$) and hierarchical clustering ($k = 2$) both looked random, while K-means ($k = 2$) showed a slight pattern.

When validating cluster assignments against manual annotations, agreement with orientation annotations was worse than random across all algorithms and cluster counts. Conversely, when comparing to machinery labelling, K-means ($k = 2$) had moderate agreement ($\text{ARI} = 0.364$).

5.3.2 Feature Group Subsetting

To identify which feature group contributes the most meaningful clustering structure, we evaluate all possible feature group subsets individually and in combination. Among all such subsets, the interperson feature group stands out, showing both the clearest structure and the strongest agreement with annotations, obtaining an ARI of 0.394 for K-means ($k = 2$), which surpasses the performance of the full feature set.

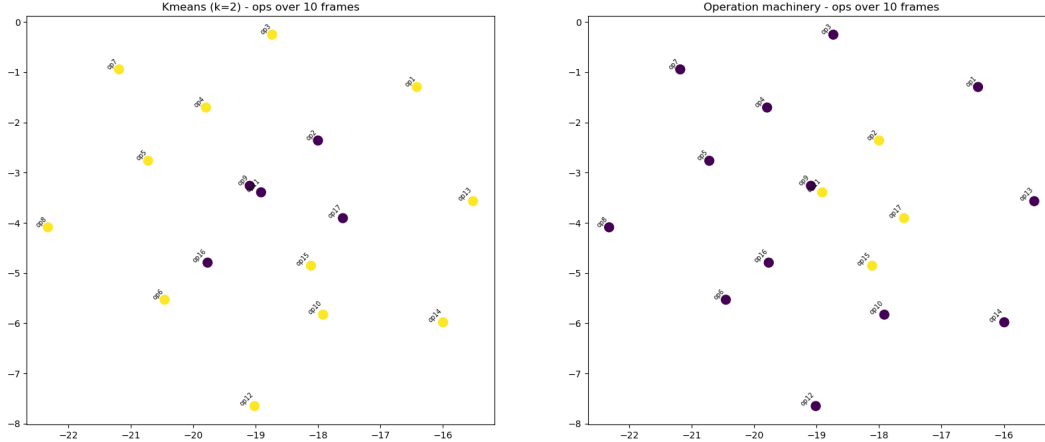


Figure 10: Machinery annotations vs. K-means ($k = 2$) with frame filter.

Supplementing the interperson group with other feature groups degrades performance rather than improving it, indicating that additional groups introduce noise. However, applying the operation length filter further improves the result above, increasing ARI to 0.555.

Table 5 below summarizes all feature subsets and their agreement with manual annotation.

Table 5: Feature group subset ARI vs. machinery labelling. Hierarchical clustering was not used, as it created disproportionate assignments with outlier(s) in one cluster and everything else in the other.

Subset	Agreement with annotations (ARI)
Interperson	0.394
GPD	-0.057
Table	0.026
Interperson (operation length filter)	0.555
Interperson + GPD	0.295
Interperson + Table	0.08
GPD + Table	0.026

ARI results so far indicate that there is no correlation between patient orientation and OR team dynamics; thus, we only validate against machinery annotation in subsequent analysis.

5.3.3 Feature Weight

In this section, we investigate alternatives to equal feature group weighing to see whether they yield better results.

No weights. Currently, we weight feature groups proportionally to their dimension to enforce equal contribution across all three subgroups. When omitting these weights, hierarchical clustering and K-means visually look random at both $k = 2$ and $k = 3$. This visual pattern is further

supported by agreement with manual annotations, which was no better than random across all algorithms and k values.

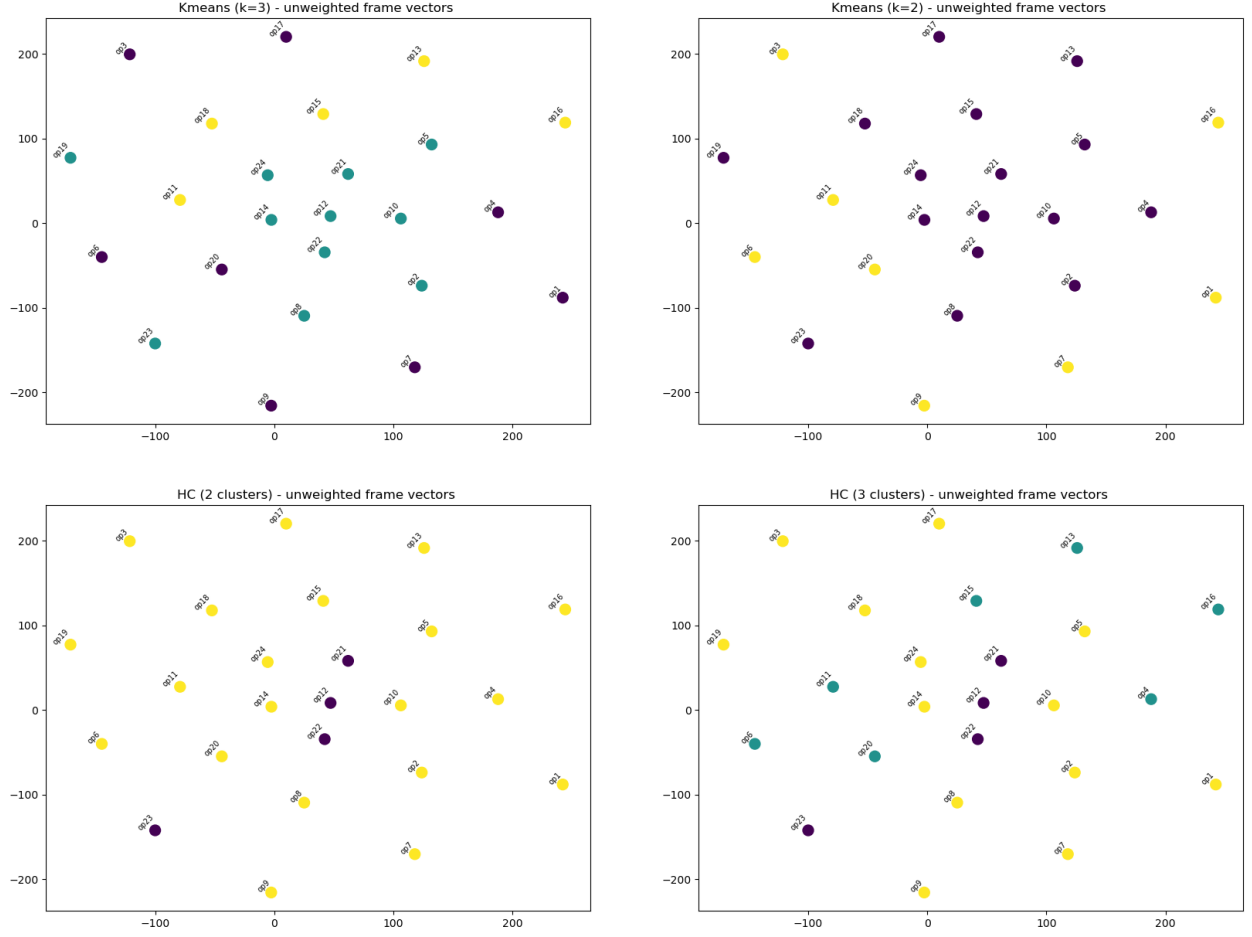


Figure 11: No feature group weight clustering results

These results are within expectation: together, GPD and table features account for 55 of the 64 total features, yet individually they both perform poorly. Therefore, by letting these high-dimensional feature groups dominate the feature space, this outcome was inevitable.

AHP weighing. Returning to the AHP approach discussed in Section 4.1.2, we derived weights using a mix of the results from the feature group ablation and our own understanding of the problem, since we had no access to domain experts. AHP produced the following weights: GPD 12.2%, Interperson 64.8%, and Table 23%. Table 6 below shows the comparison matrix that generated these weights.

The AHP weighing approach yielded similar results to the original proportionate weighing. Cluster assignment for K-means ($k = 2$) showed slight agreement with annotations, achieving an ARI of 0.21, only a negligible difference from the original approach. These results indicate that, with access to domain experts, expert-informed weighing may produce far greater results than both our AHP weighing and the proportionate weighing approach.

	Interperson	GPD	Table
Interperson	1	5	3
GPD	1/5	1	1/2
Table	1/3	2	1

Table 6: AHP pairwise comparison matrix. The numbers in this matrix represent the importance we place on each feature group. This reflects the ordering Interperson > Table > GPD

5.4 Interperson Feature Group

In Section 5.3.2, we found that the interperson feature group contributes the most meaningful clustering structure. Using this as a basis, in this section we delve deeper into the interperson group to uncover which individual feature or subgroup is driving this contribution.

5.4.1 Subgroup Breakdown: Pose Diversity vs. Person Distances

On the full frame set, pose diversity achieves an ARI of 0.39, while person distances only achieves 0.21. Looking deeper into these results, we found that pose diversity had exactly the same clustering as the full interperson feature group (ARI = 1.0), indicating it is fully responsible for the interperson group’s performance in this context.

When filtering shorter operations, the relationship is inverted: pose diversity ARI slightly decreases to 0.38, and person distance ARI sharply increases to 0.55. Notably, in this setting it is now person distances which exhibits the same clustering as the full interperson set.

In an attempt to explain this counterintuitive finding, we examine the interperson features individually.

5.4.2 Individual Feature Study

Of the nine interperson features, based on ARI, the ones that contribute the most are:

- Pose difference standard deviation: ARI = 0.29 (K-means), 0.14 (HC)
- Mean clinician distance: ARI = 0.29 (K-means), 0.14 (HC)
- Minimum clinician distance: ARI = 0.29 (K-means), 0.14 (HC)

These three features all have identical cluster assignments for both hierarchical clustering and K-means. Given low clinician counts, mean and minimum will naturally be very similar; thus, mean clinician distance and minimum clinician distance having the same results is not surprising. However, pose diversity standard deviation and mean clinician distance being the same is very interesting, because these two features capture very different things.

A correlation study between pose difference standard deviation and mean clinician distance reveals that these two variables are almost perfectly correlated, with an average correlation coefficient of 0.975 across Pearson, Spearman, and Kendall correlations. Hence, this correlation suggests that in an OR context, when people are physically close to each other, their poses also tend to be similar.

Coming back to our original inquiry about interperson subsets conditionally dominating cluster assignments, that is likely due to the fact that pose difference standard deviation and mean clinician distance, which are the top contributors in their respective subgroups, are highly correlated. Nevertheless, the reason why the person distance subgroup performs better under operation filtering remains unknown.

6 Discussion

Overall, we find that interperson features are important for modelling team strategy, as they exhibit correlation with machinery presence throughout procedures. That said, we cannot conclude that interperson features alone are sufficient to model operating room team dynamics, given that they are only representative of half of the frames in the dataset. Nevertheless, this result is still a good starting point for future attempts at understanding operating room team dynamics. For future work, a few considerations with regard to the feature set and dataset should be taken into account. In this section, we list three such considerations for both feature engineering and dataset requirements.

6.1 Feature Considerations

1. **Finding a way to supplement interperson features for one-person frames that doesn't degrade performance:** This would help address the limitation identified earlier, since interperson features are currently useless for half of the dataset.

We thought about a few ways to go about this which are worth considering. One option is to use table proximity (not the ones used in this work) as a substitute on single-clinician frames. With this option, we fall back on clinician's distance to the operating table rather than leaving our vector zero valued. Another option is to find a different single person pose configuration method that complements interperson features, unlike GPD.

2. **Better way to aggregate features:** From averaging poses across frames in the GPD group to summarizing interperson relations in the interperson group, summary statistics are used all throughout the current pipeline. In all such cases, reliance on summary statistics likely washed away a substantial amount of useful information that a better aggregation method could have preserved.

Given that operations are compared using DTW, each frame's feature vector must remain flat and of consistent length across all operations, meaning features have to be collapsed into a fixed-size representation in some way. One way to do this differently than our current approach would be to treat each frame as a fully connected graph, with clinicians and the table as nodes and pairwise distances between them as edges, and then compute graph-level descriptors from this structure. This approach still reduces features to a fixed-size vector, but captures structural properties of the formation that our current single-statistic summaries may not.

3. **A more informed feature group weighing method:** Currently, our pipeline uses a proportionate weighting scheme, where each feature group contributes based on its dimensionality, due to a lack of information needed to justify a more informed alternative. This approach could be improved upon in future work.

One direction is expert-informed AHP: consulting domain experts and relevant research, then consolidating these findings using AHP to derive feature group weights. We could also attempt to implement an automatic weight-learning method. However, as discussed earlier, these methods require labels and a larger number of operations to train, and so cannot be implemented without richer data than is currently available.

6.2 Data Considerations

1. **More operations:** With 24 usable operations, only 17 having more than 10 frames, there is not much data to cluster or validate on.
2. **Richer annotations:** The MVOR dataset was not originally made for our specific task, so it neglected annotating certain information that would prove useful for us. Examples of annotations that would prove useful are operation type, surgery success, and consistent clinician labelling.
3. **Capturing poses at a quicker and more consistent frame rate:** When looking through raw images from which frames were extracted, one finds that frames exhibit large, inconsistent temporal jumps, with some operations even cutting off midway. A quicker and more consistent frame rate would allow future work to more accurately capture and understand the development of team dynamics throughout a surgery.

With regards to data, finding a publicly available dataset as rich as described above would prove difficult due to ethical and privacy concerns. Similarly, creating our own dataset while taking these considerations into account would be equally difficult. Thus, it would be wise to tackle feature considerations first.

7 Conclusion

Team dynamics in operating rooms are hard to quantify. That said, interperson interaction seems to be the level of behaviour most indicative of surgical activity, given its correlation with machinery present during surgery. Nevertheless, to take this work further and make meaningful use of its results, improvements in terms of data and/or feature engineering are needed.

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A Appendix A

12 Joints

- head, neck, left shoulder, right shoulder, left hip, right hip, left elbow, right elbow, left wrist, right wrist, left hand, right hand ⁶

3 Planes

- Left arm: LShoulder $\rightarrow$ LElbow $\rightarrow$ LHand
- Right arm: RShoulder $\rightarrow$ RElbow $\rightarrow$ RHand
- Chest/torso: Chest (avg. of left and right shoulders) $\rightarrow$ Neck $\rightarrow$ Head

19 Lines

12 directly adjacent lines (upper body):

- Head $\rightarrow$ Neck
- Neck $\rightarrow$ LShoulder
- Neck $\rightarrow$ RShoulder
- LShoulder $\rightarrow$ LHip
- RHip $\rightarrow$ RShoulder
- LHip $\rightarrow$ RHip
- LShoulder $\rightarrow$ LElbow
- RShoulder $\rightarrow$ RElbow
- LElbow $\rightarrow$ LWrist
- RElbow $\rightarrow$ RWrist
- LWrist $\rightarrow$ LHand
- RWrist $\rightarrow$ RHand

4 endsite–joint lines:

- Head $\rightarrow$ Shoulders
- Hand $\rightarrow$ Shoulders

3 endsite/endsite lines:

- Head $\rightarrow$ LHand
- Hand $\rightarrow$ RHand
- Hand $\rightarrow$ Hand

⁶Left and right hand joints were not discussed in paper but were present in the annotations